Customer Shopping Behavior Analysis

Project Report

1. Project Overview

This project focuses on analyzing customer shopping behavior for a leading retail company using transactional data. The objective is to uncover meaningful insights into purchasing patterns, customer preferences, and behavioral trends.

The analysis aims to support business decision-making by answering the key question:
"How can the company leverage consumer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies?"

By integrating Python, SQL, and Power BI, this project provides a complete data analytics pipeline—from data preparation to visualization and strategic recommendations.

2. Dataset Summary

The dataset consists of customer transactions and includes both demographic and behavioral attributes.

- **Total Records:** 3,900 purchases
- **Total Features:** 18 columns

Key Data Categories:

- **Customer Demographics:** Age, Gender, Location, Subscription Status
- **Purchase Details:** Product Name, Category, Price, Season, Size, Color
- **Behavioral Data:** Discounts, Promo Codes, Purchase Frequency, Ratings, Shipping Type

Data Quality:

- Missing values identified in the **Review Rating** column (37 entries), which were handled during preprocessing.

3. Data Preparation & Modeling (Python)

Data preprocessing and transformation were carried out using Python to ensure accuracy and consistency.

Key Steps:

- **Data Loading:** Imported dataset using *pandas*
- **Exploratory Analysis:**
  - Used .info() and .describe() to understand structure and statistics
- **Handling Missing Values:**
  - Filled missing **Review Ratings** using the **median value per product category**
- **Data Cleaning & Standardization:**
  - Renamed columns into **snake_case** format for consistency
- **Feature Engineering:**
  - Created **age_group** by categorizing customer ages
  - Generated **purchase_frequency_days** to analyze buying behavior
- **Redundancy Removal:**
  - Removed promo_code_used after identifying overlap with discount_applied
- **Database Integration:**
  - Loaded the cleaned dataset into **PostgreSQL** for further analysis

4. Data Analysis (SQL)

SQL was used to simulate business queries and extract actionable insights.(used PostgreSQL)

Key Analyses Performed:

1. **Revenue by Gender** - Compared total spending between male and female customers

	gender text	revenue numeric
1	Female	75191
2	Male	157890

2. **High-Spending Discount Users** - Identified customers who spend above average even when using discounts

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
11	24	88
12	29	94
13	32	79
14	33	67
15	35	91
16	37	69
17	40	60
18	41	76

Total rows: 839 Query complete 00:00:0

3. **Top 5 Products by Rating** - Highlighted highest-rated products based on customer reviews

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

4. **Shipping Type Analysis** - Compared spending behavior between **Standard** and **Express** shipping

	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

5. **Subscribers vs Non-Subscribers** - Analyzed revenue and average spending across subscription groups

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

6. **Discount-Driven Products** - Identified products most frequently purchased with discounts

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.66
3	Coat	49.07
4	Sweater	48.17
5	Pants	47.37

7. **Customer Segmentation**
   - Classified customers into: New, Returning, Loyal

	customer_segment text	Number of Customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

8. **Top Products per Category** - Extracted top 3 most purchased products in each category

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessori...	Jewelry	171
2	2	Accessori...	Sunglasses	161
3	3	Accessori...	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

9. **Repeat Buyers & Subscription Behavior** - Examined whether frequent buyers are more likely to subscribe

	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

10. **Revenue by Age Group** - Determined which age groups contribute most to revenue

	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

5. Visualization & Dashboard (Power BI)

An interactive Power BI dashboard was developed to present insights in a clear and engaging way.

Dashboard Highlights:

- Sales trends across categories and seasons
- Customer segmentation and demographics
- Impact of discounts and ratings on purchases
- Subscription vs non-subscription performance
- Shipping preferences and spending behavior

Key Features:

- Interactive filters (Age group, Category, Gender, Subscription)
- KPI cards for revenue and customer metrics
- Visual comparisons using bar charts, pie charts, and line graphs

This dashboard enables stakeholders to quickly explore patterns and make data-driven decisions.

6. Business Recommendations

Based on the analysis, the following strategic recommendations are proposed:

1. Enhance Subscription Programs

- Offer exclusive discounts and benefits to increase subscriptions

2. Strengthen Customer Loyalty

- Introduce reward systems for repeat customers to move them into the **loyal segment**

3. Optimize Discount Strategy

- Balance discount usage to maximize revenue without reducing profit margins

4. Promote High-Performing Products

- Focus marketing efforts on **top-rated and best-selling products**

5. Targeted Marketing Campaigns

- Personalize campaigns based on:
  - High-spending age groups
  - Frequent buyers
  - Express shipping users

7. Conclusion

This project demonstrates how combining **Python, SQL, and Power BI** can transform raw data into actionable business insights. By understanding customer behavior, the company can improve engagement, optimize marketing strategies, and drive long-term growth.